Planted for a market that no longer exists

A forest, a village, and a sideways idea:
history, today, and what we are still working out



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Kinjo Rie (Mitsue Kanko) · James Hill · Rob Oudendijk (Mitsue-it)

Three people, one forest, some trees very close to the front door

Kinjo Rie

Mitsue Kanko. Lives and works in the village; knows what the forest looks like on a Tuesday, not just in a plan.

Takes the middle of the talk: the situation on the ground today.

James Hill

Forest history researcher. Walks us through how Japan ended up with this much sugi, and why.

Takes the first part of the talk.

Rob Oudendijk

Dutch engineer. Built his own house in the Mitsue forest, so the trees are 3 m from the front door.

If the talk goes badly, blame the neighbours.

Dress-code idea (pending Kinjo-san's OK): she in green, new growth; he in grey, fully grown. Together: one forest.

History, today, a sideways idea, and what we have not figured out yet

1	**History**	Japanese forestry, from the Tokugawa to 1958	James
2	**Today**	The forest and village as they are now	Kinjo-san
3	**The problem**	Three problems with one shape	Rob
4	**The idea**	Pull, don't push: wood, power, replanting	Rob
5	**Homework**	What we checked, and what we still do not know	All

Tone: light, not academic. Big scary problems sometimes have weirdly simple solutions, if you look at them sideways.

JAMES · TO EXPAND

Three centuries in one strip: how Japan came to plant so much sugi

1603–1867

Tokugawa

Land is feudal or communal. Edo and Osaka grow huge and burn and rebuild; planted timber starts to pay.

1868–1947

Meiji to the war

Forests are split between state and village; the Imperial estate runs forestry as a real business.

1929–1949

Cutting outruns planting

Twenty years of overcutting, then wartime emergency harvest.

1950–1958

Replant, replant, replant

Compulsory planting, three national plans, the biggest boom ever.

Why it matters here

Four chapters, ten slides.

The punchline arrives in 1958: the trees we see today were planted just as the demand behind them began to fade.

Sources: Totman; Iwamoto; Matsushita (2015)

JAMES · TO EXPAND

Who owns a tree on shared land? The question that made plantations possible

1603–1867

Tenure

Land is held by feudal lords or villages in common; private land ownership is banned in principle. Lord's forests in Kiso are guarded and theft is severely punished.

1665 · 1724

Kiso reforms

1665: cutting halved by seedling protection and patrols, reversed within years for revenue.
1724: cut down by more than 60%, and this time held for 30 years.

late 1600s

Wave 1

Edo and Osaka boom; a 50-year timber investment pays for the first time. Local custom starts to recognise private ownership of planted trees even on communal land (Yoshino, Owase).

1801

Edo

Population of Edo around 1,000,000, bigger than London or Paris. With imports shut out, all demand falls on domestic forests.

Why it matters here

Plant a tree on land you do not own, and who gets the timber 50 years later?

The answer worked out in Yoshino and Owase, in the 1600s, is the same question that sits behind Mitsue's private woodlots now.

Sources: Iwamoto; Totman, Green Archipelago

JAMES · TO EXPAND

Edo-era foresters already had a recipe: plant dense, thin early, thin again

1750s

Yamaguni guild adapts

A rural timber guild creates a new licence class so plantation-growers can join without losing collective control. Rural producers start out-organising city merchants.

1779

Kiso inventory

First forest inventory and planning: early sustained-yield thinking.

1849

The Yoshino method, written down

Okino Takao (Kurobane domain) documents: plant at 3 ft, thin at 2–3 inch butt, rethin at 5 years, final spacing 12 ft. A citable schedule that still looks familiar.

Over 250 years later

Plantations of this kind are what stand on Mitsue's slopes.

Why it matters here

The thinning rules are old. What is new is that there is nobody to do the thinning.

Good hook into Kinjo-san's segment on the day-to-day work now.

Sources: Totman; Iwamoto

JAMES · TO EXPAND

Meiji tries to divide every woodland into 'government' or 'private'

1868

Meiji Restoration

Formal land ownership is introduced. National forests are carved out of communal and former feudal land.

1871–76

Six policy reversals in five years

Villagers need gen'ya, mixed scrub for fodder and fuel. The state wants sanrin, dense conifer. A third category, koyuchi (shared use), appears because the first two do not fit real life.

1889

Result

Government forest reaches 53% of Japan's woodland (31% in 1880). Messy: use-rights linger on both sides.

Why it matters here

The same tension in a Japanese village: what the state wants from the forest is not what the village needs.

Worth a joke: six policy reversals in five years is a pace most of us would not call 'planning'.

Sources: Totman, 'Green Archipelago'

JAMES · TO EXPAND

For 58 years, forestry was a real business: the Imperial forests

1889–90

Founded

About 1.6 million ha carved out as the Imperial Household's own income-producing forest estate.

1892

Turns profitable

After one or two loss-making years: the start of a real sustained-yield P&L.

1936–45

Peak

Around 40% net margin on gross income, against 23% for the much larger National Forest. 31% of Household income in 1940.

1942–47

Override and end

'Hundred Year Plan' suspended in wartime; harvest up by cutting replanting. Dissolved 1947; asset value had reached ¥2.6bn, 70% of Household wealth.

Why it matters here

Managed Japanese forestry was, for decades, a real bankable asset and not a subsidised cost.

Our own slogan 'Grow the Forest. Grow the Village.' comes from a hundred-year plan like this.

Sources: Totman, Goryōrin and lumber industry notes

JAMES · TO EXPAND

Twenty years in a row, more forest was cut than planted

1929–49

The gap

Annual cut area exceeds planted area nationally every year for two decades; the gap peaks in the mid-1940s.

1942–44

Wartime override

'Special harvesting' authorised: harvest rises from 8.6 to 10.97 million koku, paid for by cutting reforestation and aftercare.

1948

Cutover land

1.49 million ha, around 6% of Japan's forest area, concentrated near villages.

1948

The forecast

GHQ and Forestry Agency studies predict domestic wood will run out within 15–33 years (wrong, in hindsight, but it drove everything that followed).

Why it matters here

A scarcity scare, followed by a planting rush.

Many of Mitsue's stands are the direct descendants of that rush.

Sources: Matsushita (2015); Totman

JAMES · TO EXPAND

Three plans in twelve years, and planting becomes compulsory

1949–53

5-Year Plan

First full postwar plan for 1.5 million ha of cutover land; budget shortfalls keep execution at 60–70%.

1951

Forest Act amended

Planting made compulsory for the first time (Article 14). Same year: log import tariffs cut to near zero.

1952–61

10-Year Plan

Self-sufficiency in industrial roundwood through broadleaf-to-conifer conversion (0.84 million ha planned).

1955–60

6-Year Plan

From quantity to quality: nursery stock, seed trees, 'right tree, right site'.

Why it matters here

Broadleaf-to-conifer conversion is exactly what we now want to undo, in places where it makes sense.

A tidy irony for the talk: the plan of the 1950s and the plan of the 2020s point in opposite directions.

Sources: Matsushita (2015)

JAMES · TO EXPAND

Just as planting peaks, the reason for planting starts to disappear

1950

The demand floor drops out

Fossil fuels replace firewood and charcoal, chemical fertiliser replaces green fertiliser, machinery replaces draft animals. Village woodland (gen'ya) starts to lose its value.

1954

Peak planting year

Highest annual planted area recorded under the 10-Year Plan.

1955–1970

Wave 3, the biggest

Postwar recovery drives the largest of Japan's three plantation booms. Artificial planting peaks in 1955 and stays high until 1970.

Why it matters here

Planting goes up. Demand for what the trees were meant to supply goes down.

(Idea for James: a simple two-line chart here, 'planted area' vs 'firewood and charcoal use'. Schematic is fine if labelled.)

Sources: Totman; Iwamoto; Matsushita

JAMES · TO EXPAND

Two quiet decisions that explain the next 70 years

1951

Import tariffs cut

Log import tariffs lowered to near zero. At the time it looked sensible; it set up self-sufficiency to fall for decades.

1956

Forest Development Corporation

A national vehicle, with prefectural corporations from 1959, plants the least attractive land with long-term debt against a 40–50-year harvest. All 47 prefectural corporations end up in debt distress once stumpage prices collapse after 1980.

1970s–80s

Cheap imports

The market crashes before the planted trees grow up. A-grade sugi falls from about ¥12,000 to ¥4,475/m³ by 2025.

Why it matters here

We planted a retirement fund, then the market crashed before the trees were old enough to cash in.

Only A-grade wood sells today. That is the problem the later part of the talk tries to solve.

Sources: Matsushita (2015); price series to be confirmed

JAMES · TO EXPAND

1958: Rob's birth year, and the year the story turns

1958

Deep in the biggest boom

Planting compulsory since 1951; self-sufficiency still about 95% and about to start its long decline.

1958

A new financing model

The Profit-Sharing Forestation Act follows the same year; debt-financed corporate planting is formalised.

today

What is standing now

The sugi and hinoki in Mitsue's mountains were planted in exactly this window.

Why it matters here

The trees are older than the demand that justified planting them.

That is the hinge into 'today'. Over to Kinjo-san.

Sources: Matsushita (2015); Iwamoto; Rob's own notes

The forest as it is now: Kinjo-san's slides go here

Placeholder. Kinjo-san decides what she wants to show: the day-to-day, photos of the forest, what the work looks like.

Ideas to offer her, not requirements:

- A few photos of the stands: thinned, unthinned, left-in-forest wood
- What the cooperative does, and what it cannot do at the moment
- What the village hall plan says about thinning (see next slides)
- What she would like NAIST students to look at

Sources: to be written with Kinjo-san

STILL RESEARCHING

A village of about 1,400 people, and a forest of about 7,000 hectares

≈ 1,400	≈ 93 %	≈ 7,050 ha	50–60 yr
residents	forest cover	forest, over 90% private	age of the main sugi and hinoki stands

Most of the forest is private, many owners are elderly or live elsewhere, and a lot of it was planted in the 1950s–60s boom you just saw.

At today's prices a cut sugi barely pays for the cutting. Only the straight, clean A-grade logs sell, and even that price has fallen by about two thirds since the 1980s.

Still checking: how many owners, how many parcels, who inherits. The forest register is with the village hall.

Sources: Village forestry plan; Mitsue statistics; draft outline

Three problems with one shape: nobody is paid to fix it

1 Not enough hands

The cooperative is small and has few young people. Harvest at scale needs machines and trained operators; we have neither yet.

Thinning is planned, but most of it is cut and left.

2 A ticking clock

A large share of the forest reaches harvest age inside the same ~25-year window.

The same window in which new small-scale power technology matures. Nice timing, if we use it.

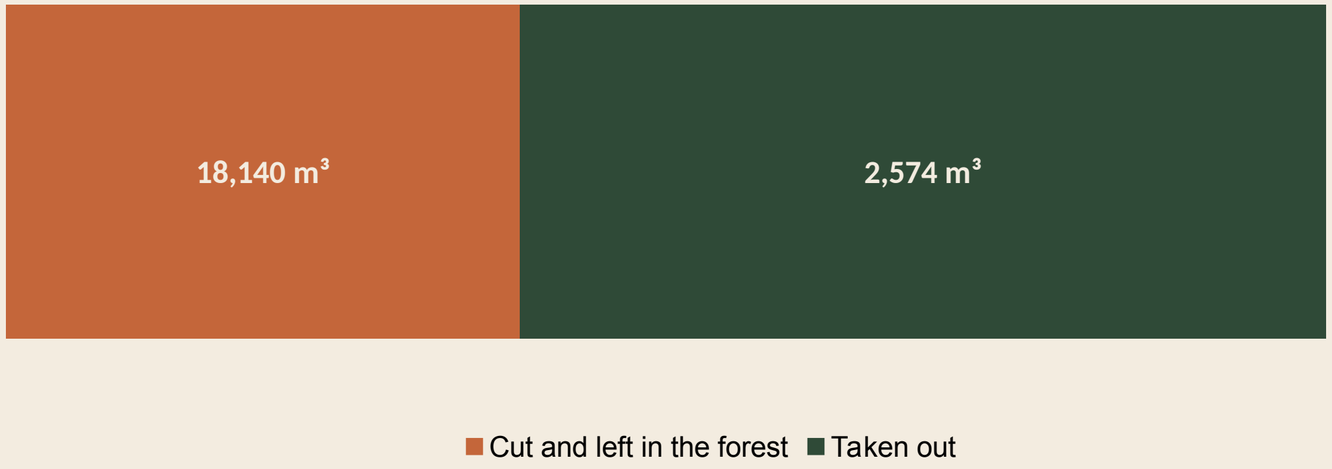
3 No pull to replant

Cutting sugi for cheap lumber does not pay enough to put anything back.

Short-term profit, repeat forever. Nobody has a reason to break the cycle.

STILL RESEARCHING

Most of the thinning wood stays in the forest



≈ 88 % stays put

113 parcels, about 222 ha, in 4 districts.

About 20,700 m³ planned; about 860 m³ a year actually comes out.

That cut-and-left wood is a fuel resource, if it can be collected at a price that makes sense. Steep slopes and missing roads are why it is left.

Still checking: how much of it is reachable with a winch-assisted machine, and what an earlier '120–300 m³/yr' cooperative figure really measured.

Sources: Village forestry plan R6–R8 (FY2024–26); Oba, village hall

Pull, don't push: give the wood a buyer that pays

1 Thin and clear

Cut-to-length machines on steep sugi

›

2 Sort A / B / C / D

A and B to the sawmill; C and D to chips

›

3 Chip and dry

Local chipping, drying with CHP heat

›

4 Small CHP

Gasification plant: power and heat, 24 h

Why 'pull'.

Waiting for a subsidy pushes wood out of the forest once, then stops. A buyer that wants fuel 24 hours a day pulls wood out every year, and can pay for the replanting.

Then replant.

Cleared stands go back to native broadleaf (oak, chestnut, konara) for wildlife and watershed, with a fast-growing kiri (paulownia) fuel crop on part of the land. Species are chosen with the NGO more trees.

STILL RESEARCHING

Only A-grade wood sells today. B, C and D are a problem we can use

A Straight, clean, no knots	**B** Slightly bent, some knots	**C** Crooked, small	**D** Branches, tops, bark
Sawlog. The only grade worth cutting for lumber.	Sawlog or laminated; price depends on buyer.	Chips, pulp, board. Cheap.	Today: left on the slope.

CHP fuel and a data-center customer mean C and D wood suddenly has a buyer too. The sawlogs still fund most of the work.

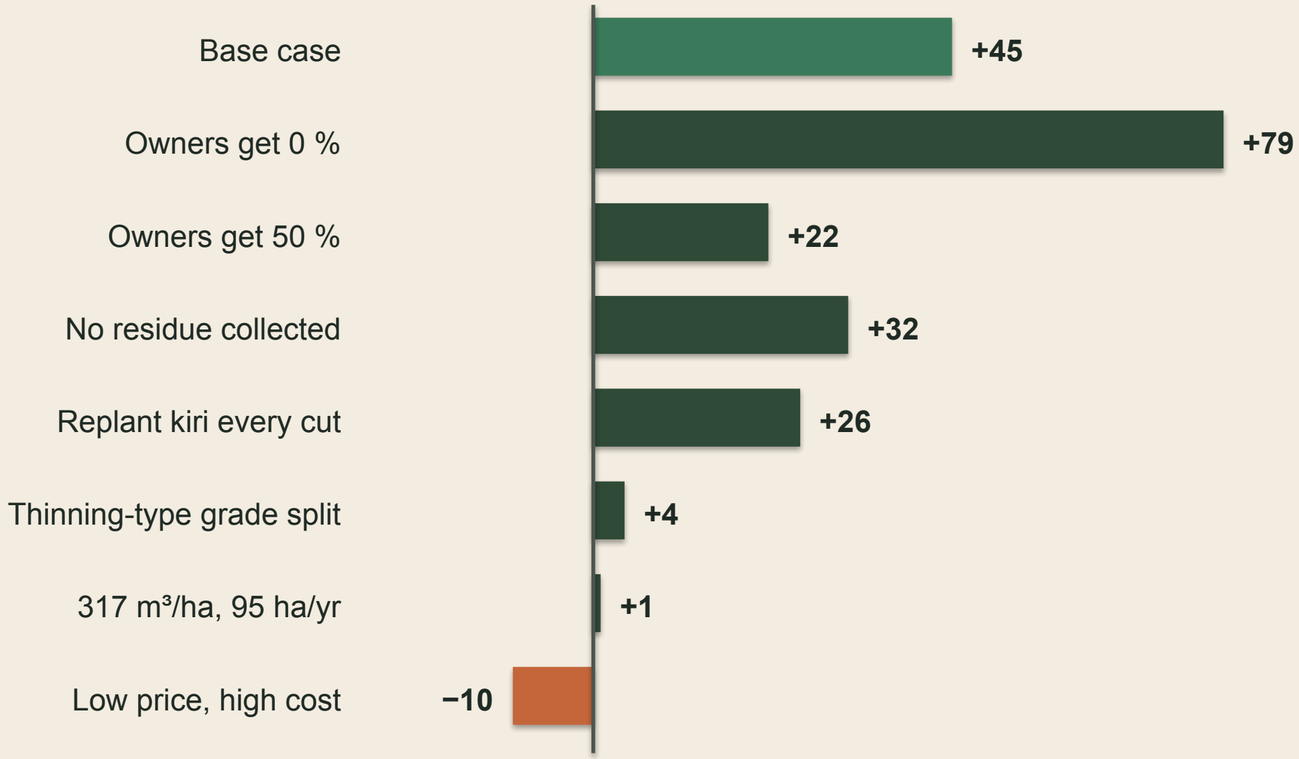
Still checking: what share of Mitsue's neglected stands is really A-grade. The national average is about 61%; thinning-type stands can be near 22%.

Sources: Forestry Agency grade conventions; Japanese-wood market notes

PLACEHOLDER PRICES

Our first model says plus, and then shows how fragile that is

Steady-state system net, ¥M per year (placeholder prices)



Not a forecast

50 ha/yr clear-cut at 600 m³/ha, sawlogs sold, C+D chipped, half the land replanted to kiri, landowners paid 30% of sawlog net.

At the village's actual standing volume (277–317 m³/ha) the base case falls to about break-even.

Left out: carbon credits, data-center income, machine purchase, wages.

Sources: kiri_sugi_integrated_system_model_2026-11-03.xlsx v1.2. All prices and costs are placeholders, not quotes.

STILL RESEARCHING

Four unknowns decide the answer, not the technology

1 Wood per hectare

Model uses 600 m³/ha. The village plan shows 277–317 m³/ha standing.

2 A-grade share

Sawlog sales fund everything. National average 61%; Mitsue may be much lower.

3 Landowner share

Each 10 points to owners costs about ¥11M a year. We chose 30% as an incentive.

4 Residue price

Placeholder ¥4,000/t. Doubling it adds only about ¥1.6M a year: minor.

Ordered by how much they move the result. Sources and calculations in docs/energy-forest/economics/.

STILL RESEARCHING

Start small to learn, size the real plant to the wood we can actually get

Step 1: a 50 kWe prototype

Small gasification unit in a 20 ft container. Needs about 450–600 t of dry wood a year for 24 h running.

A learning step, not an economic unit: efficiency only about 12–20% at that size.

Site not decided.

Step 2: about 0.5–0.6 MWe

Our current view of the first commercial size. Efficiency 23–28% for gasifier plus engine at that scale.

A 1.2 MWe plant needs about 24,000 m³ of fuel a year; we would not size for it until the harvest is proven.

At 1,000 kW and above the tariff is FIP-only, and splitting is not allowed.

Three more constraints

Japanese maker only. Gasification plus engine beats steam below about 1 MWe.

Grid: the local substation has about 3 MW locally and none upstream. Prior consultation drafted, not filed.

Fuel may be bought from neighbouring villages too, not only from Mitsue.

Sources: Energy, finance and resources overview v1.1 (2026-11-03)

STILL RESEARCHING

Pieces of this already exist, in Japan and abroad

Tenkawa, next door

A biomass council since 2016. A wood boiler heats the Dorogawa hot spring baths. Photos from our visit on 25 Aug 2026.

Aizu, 2011

A closed school (Akazawa Elementary) turned into a data center. A precedent for what to do with an empty school building.

Wood power plus compute

Biomass-powered or biogas-powered compute has been tried: Buena Vista (US), Deutsche Telekom with Bankhaus Metzler (Germany), IIJ Matsue (Japan).

Still checking: what exactly each of these is, and what actually transfers. Treat each as a lead, not as proof.

Sources: Tenkawa Forest Power Council; Aizu (2011) Safecast blog; Buena Vista, DT/Metzler, IIJ Matsue

The homework so far

Read the law

45 village, prefectural and national acts on forestry, biomass, 協力隊 and siting collected and summarised.

Read the village

Council minutes and the village newsletter archive, run through OCR; the thinning plan for 113 parcels.

Asked the makers

Two rounds with Komatsu on steep-slope harvesting; gasification CHP makers compared; FFPRI references.

Built a model

A spreadsheet linking clear-cut, sawlogs, chips, kiri, CHP and landowner payments. Version 1.2.

Went to the sites

Forest visits, the village hall, the Uda City cooperative and the Tenkawa meeting.

Wrote it down

Everything is open in a public repository, so anyone, including NAIST students, can check or challenge it.

Sources: repository docs/, research/, docs/reference/

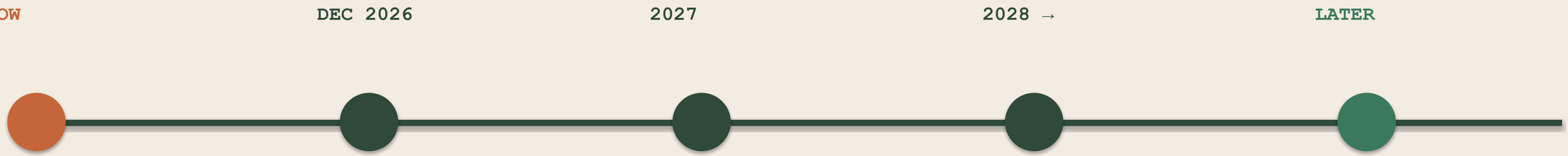
STILL RESEARCHING

Honest list of open questions, and who is answering them

OPEN QUESTION	WHO	STATUS
Stand volume, stumpage, who gets the sale income	**Village hall (Oba)**	asked 6 Nov
Slope limits with and without winch, road width, yearly output	**Komatsu**	follow-up sent
Real CHP efficiency and cost at 0.6–0.8 MWe	**Maker quotes**	to request
Kiri growth on Mitsue slopes	**SANSEI planting data**	asked 2 Nov
Grid connection at 6.6 kV	**Kansai T&D**	consultation drafted
Fuel bought in from neighbouring villages	**Uda City, others**	to approach
What students and researchers would measure	**NAIST (you)**	to discuss

Sources: docs/strategy/mitsue_energy_finance_resources_overview_2026-11-03.md section 5

Early planning. No money raised, no contracts, no machines bought



Phase 0

Trust, sites, fuel talks, homework

Phase 0 ends

Decision on first site and partners

Prototype

50 kWe CHP and fuel trial, if site and partners are settled

Machines

A CTL trial, if a fit is shown and financing is in place

Full scale

About 0.5–1.2 MWe and a larger compute load; no fixed date

Status: ¥0 committed. Dates are targets, not promises. The grant tracks we know of (METI GX regional, MoE zero-emission data center) are being studied, not applied for.

75 years to grow the forest.

25 years to grow the village.

A hundred-year plan in the old Goryōrin spirit: planting for the people who will be here when the trees are ready.

Tell us where we are wrong

Honest critique is more useful to us than polite interest.

What would you measure? What would you doubt? Who else should we ask?

Rob Oudendijk · Mitsue-it · rob@mitsue.it

All documents open: https://codeberg.org/YR-Design/mitsue-ai-data-center

Website: mitsue.it